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MILITARY SPECIFICATION

NIGHT VISION GOGGLES AN/PVS-5A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Night Vision Goggles, AN/PVS-5A (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	- Pallet, Materials, Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
QQ-S-781	- Strapping, Steel, Flat and Seals
PPP-B-585	- Boxes, Wood, Wirebound
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock Corner
PPP-B-636	- Boxes, Shipping Fiberboard
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	- Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-S-760	- Strapping, Nonmetallic (and Connectors)
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing)
PPP-T-97	- Tape, Packaging/Industrial, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, USA CECOM, ATTN: DRSEL-RD-TS-S, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-B-18	- Batteries, Dry
MIL-P-116	- Preservation-Packaging, Methods of
MIL-P-11268	- Parts, Materials, and Processes Used in Electronic Equipment
MIL-P-19644	- Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	- Polyurethane Foam, Rigid or Flexible, for Packaging
MIL-I-49052	- Image Intensifier Assembly, 18 Millimeter Micro-Channel Wafer
MIL-L-49156	- Lens Assembly, Eyepiece, Night Vision Goggles, AN/PVS-5A
MIL-L-49157	- Lens Assembly, Objective, Night Vision Goggles, AN/PVS-5A

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Table for Inspection by Attributes
MIL-STD-129	- Marking for Shipping and Storage
MIL-STD-150	- Photographic Lenses
MIL-STD-171	- Finishing of Metal and Wood Surfaces
MIL-STD-454	- Standard General Requirements for Electrical Component Parts
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-810	- Environmental Test Methods
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment

DRAWINGS

ELECTRONICS COMMAND

SM-DL-657400	- Mask Assembly, Face
SM-DL-657420	- Adapter Assembly, Arctic
SM-DL-657430	- Case Assembly, Carrying
SM-DL-657440	- Case Assembly, Shipping and Storage
SM-DL-657443	- Light Emitting Diode Assembly
SM-DL-657500	- Night Vision Goggles, AN/PVS-5A
SM-D-657302	- Cushion Assembly Face
SM-D-657337	- Frame Assembly
SM-C-657428	- Demist Shield Assembly

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The Night Vision Goggles, AN/PVS-5A consists of a goggle assembly, a battery, a carrying case, a shipping/storage case, and ancillary equipment. The goggle assembly hereinafter referred to as the assembly, is a lightweight, headmounted, self-contained Night Vision System containing two monocular units comprising a binocular system. Each monocular unit consists of an objective lens assembly, an image intensifier tube assembly, and an eyepiece assembly. The monocular units are mounted in an aluminum frame assembly. The frame is mounted to a face mask assembly which is held by head straps to the user's head. The assembly incorporates an infrared (IR) light source which provides illumination, when required, to permit close-in-viewing. The assembly permits the user to view at ranges of ten inches to infinity by means of a focus adjustment.

3.2 Construction. The assembly shall be in accordance with SM-DL-657500 and all provisions of this specification.

3.3 First article. The supplier shall furnish and test assemblies as specified in the contract or purchase order (see 6.2).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268, with the exception of the image intensifier assembly whose materials, parts and processes shall conform to MIL-I-49052.

3.5 Major components. The following major components shall be constructed in accordance with applicable drawings, specifications, and as specified herein.

3.5.1 Arctic adapter assembly. The arctic adapter assembly shall be in accordance with Drawing List SM-DL-657420 and as specified herein.

3.5.2 Demist shield assembly. The demist shield assembly shall be in accordance with Drawing SM-C-657428.

3.5.2.1 Fogging. Fogging shall not occur on the shield over a temperature range from $0^{\circ}C$ to plus $45^{\circ}C$ and relative humidity of 95 percent, ± 5 percent.

3.5.2.2 Damage. The shield shall not be damaged in a relative humidity of 95 percent, ± 5 percent over a temperature range from $-54^{\circ}C$ to $+65^{\circ}C$.

3.5.3 Lens assembly, objective. The lens assembly, objective shall be in accordance with MIL-L-49157.

3.5.4 Lens assembly, eyepiece. The lens assembly, eyepiece shall be in accordance with MIL-L-49156.

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3.5.5 Image intensifier assembly. The image intensifier assembly shall be in accordance with MIL-I-49052.

3.5.6 Frame assembly. The frame assembly shall be in accordance with Drawing SM-D-657337 and as specified herein.

3.5.7 Face mask assembly. The face mask assembly shall be in accordance with Drawing List SM-DL-657400 and as specified herein.

3.5.8 Face cushion assembly. The face cushion assembly shall be in accordance with Drawing SM-D-657302 and as specified herein.

3.5.9 Light emitting diode assembly. The light emitting diode assembly shall be in accordance with Drawing List SM-DL-657443 and as specified herein.

3.5.10 Case assembly, carrying. The case assembly, carrying shall be in accordance with Drawing List SM-DL-657430 and as specified herein.

3.5.11 Case Assembly, shipping and storage. The case assembly, shipping and storage shall be in accordance with Drawing List SM-DL-657440 and as specified herein.

3.6 Operational and environmental.

3.6.1 Controls and adjustments.

3.6.1.1 Rotary switch. The rotary switch shall rotate to 3 distinct positions. In the OFF position neither one of the image intensifier tubes nor the IR diode shall light. In the ON position both of the image intensifier tubes shall light and the diode shall not light. In the IR position both of the tubes shall light and the IR diode shall light.

3.6.1.2 Interpupilary adjustment. With the lever, clamp tightened, both monocular assemblies shall be held so as to prevent loss of adjustment. With the lever, clamp loosened, monoculars shall be capable of movement throughout their entire interpupilary adjustment range.

3.6.1.3 Relative movement of binocular assembly. The binocular assembly shall be capable of movement relative to the face mask when the two knob clamps are loosened. The nut, self locking hex, shall permit the knob clamp to be loosened while at the same time retaining it on the screw, shouldered.

3.6.2 Resolution.

3.6.2.1 Far focus. Each monocular assembly shall have an on-axis resolution at the infinity stop of not less than 20 lp/mm or 5 lp/mm less than the maximum tube resolution of the monocular, whichever is greater, and shall have an on-axis resolution at the true infinity setting of not less than 23 lp/mm, or 4 lp/mm less than the maximum tube resolution of each monocular, whichever is greater.

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3.6.2.2 Close focus. Each monocular assembly shall have an on-axis resolution at the close focus stop of not less than 23 lp/mm for a target at a distance of not more than 25 centimeters (cm).

3.6.3 Brightness gain. The brightness gain shall be not less than 400 footlamberts per footlambert. There shall be no change in brightness gain greater than 3 percent of the average gain when the assembly is subjected to steady input illumination.

3.6.4 Collimation. When two parallel beams of collimated light are projected into the objectives, with the objectives adjusted to and operating in the infinity focus position and both eyepieces set at zero diopter, the conjugate beams emerging from the two eyepieces shall be collimated within 1° divergence/convergence in the horizontal plane and within $1/2^{o}$ in the vertical plane.

3.6.5 Focus adjustment. The torque required to adjust the focus range shall be 25 inch-ounces, ±15 inch-ounces.

3.6.6 Diopter focus. The minimum diopter focus range shall be from +2 to -6 at a 15 mm eye relief. The torque required to adjust the eyepiece diopter focus shall be not less than 10 inch-ounces nor greater than 105 inch-ounces.

3.6.7 Vibration. The assembly shall operate during, and not be damaged by vibrations in the frequency range of 10 to 55 Hertz (Hz). In addition, the assembly in its carrying case, and in turn, in the shipping case shall be capable of withstanding the vibrations normally induced during transportation as loose cargo without damage or degradation of performance.

3.6.8 Shock. The assembly in its carrying case and in turn, in the shipping case, shall not be damaged and shall operate without degradation or performance after being dropped from a height of not less than 48 inches.

3.6.9 Temperature, humidity, altitude. The assembly shall withstand storage at temperatures ranging from $-57^{o}C$ to $+65^{o}C$, altitudes ranging from sea level to 50,000 feet above sea level while at a temperature of $-57^{o}C$, and a humidity level up to 95 percent relative humidity at temperatures $+65^{o}C$. The assembly shall operate at temperatures ranging from $-45^{o}C$ to $+45^{o}C$. The assembly shall withstand temperatures ranging between $+65^{o}C$ and $-57^{o}C$ within a five minute period.

3.6.10 Immersion. The assembly shall not be damaged and shall operate without degradation of performance after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes.

3.6.11 Electromagnetic interference (EMI). The assembly shall meet the requirements of MIL-STD-461, Class 1C. The required tests are RE02, RS01, RS02, and RS03 only.

3.7 Treatment and painting. Unless otherwise specified the metallic parts of the assembly shall be given a treatment and painting per MIL-STD-171.

3.8 Marking. The assembly and all components, subassemblies, and parts shall be marked in accordance with SM-DL-657500.

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3.9 Workmanship. The assembly shall conform to Requirement 9, MIL-STD-454. Joints and seams shall be a tight fit, and electrical wiring shall be secure with unbroken insulation. Padding shall be free of gouges, tears, excess adhesive, or imbedded foreign matter and shall be attached firmly to the proper surface.

3.10 Reliability. The assembly shall have a specified Mean Time To Failure (MTTF) of 2000 hours when operated under the reliability test conditions specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Inprocess inspection (4.3).
b. First article inspection (4.4).
c. Quality conformance inspection (4.5).
d. Inspection of packaging (4.10).

4.3 Inprocess inspection. Prior to assembly, each component listed in Table I shall be inspected in accordance with Table I. An inprocess inspection lot for demist shields shall be defined as an individual shipment received for incoming inspection. Samples shall be selected in accordance with MIL-STD-105, general inspection level II, AQL of 1.5 percent.

TABLE I. Inprocess inspection.

Inspection	Requirement paragraph	Test paragraph
Demist shield assembly:		
Fogging	3.5.2.1	4.7.2.1
Damage	3.5.2.2	4.7.2.2

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4.4 First article inspection.

4.4.1 Inspection. Each first article assembly shall be inspected in accordance with Table II, either in process or when assembled, as required. Presence of one or more defects shall be cause for rejection of that assembly and may be cause for rejection of first article.

TABLE II. Inspections.

Defects	Requirement paragraph	Inprocess	Assembled
Parts, materials, and components missing or not as specified	3.2, 3.4	X	X
Identification or special marking missing or illegible	3.2, 3.8	X	X
Workmanship	3.9	X	X

4.4.2 Tests. Following successful completion of the inspections specified in 4.4.1, four first article assemblies with intensifiers which have met MIL-I-49052, objectives which have met MIL-L-49157, and eyepieces which have met MIL-L-49156 shall be subjected to all the inspections in Table III except for reliability. Five first article assemblies, with intensifiers which have met MIL-I-49052, objectives which have met MIL-L-49157, and eyepieces which have met MIL-L-49156, and which have met inspections listed in Tables IV and V shall be subjected to the reliability inspection only. Inspections listed in Table III may be performed in any order except resolution shall be performed last. Failure of any inspection shall be cause for rejection of that assembly and may be cause for rejection of first article.

TABLE III. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Arctic adapter assembly:	3.5.1	4.7.1
Demist shield assembly:		
Fogging	3.5.2.1	4.7.2.1
Damage	3.5.2.2	4.7.2.2
Operational and environmental:		
Controls and adjustments	3.6.1	4.8.1
Brightness gain	3.6.3	4.8.3
Collimation	3.6.4	4.8.4
Focus adjustment	3.6.5	4.8.5
Diopter focus	3.6.6	4.8.6
Vibration	3.6.7	4.8.7

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TABLE III. First article inspection. (Cont)

Inspection	Requirement paragraph	Test paragraph
Shock	3.6.8	4.8.8
Temperature, humidity, altitude	3.6.9	4.8.9
Immersion	3.6.10	4.8.10
Electromagnetic interference	3.6.11	4.8.11
Resolution	3.6.2	4.8.2
Reliability	3.10	4.9

4.5 Quality conformance inspection.

4.5.1 Group A inspections. Initial inspections shall include the Night Vision Goggle, SM-D-657300 only. Final examination shall include all items of the Night Vision Goggle, AN/PVS-5A, SM-D-657500. Each AN/PVS-5A shall be inspected in accordance with Table II. Each assembly shall be inspected in accordance with Table IV. Failure of an AN/PVS-5A to pass the inspections of Table II shall be cause for rejection of that AN/PVS-5A. Failure of an assembly to pass the inspections of Table IV shall be cause for rejection of that assembly.

TABLE IV. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
Controls and adjustments	3.6.1	4.8.1
Resolution	3.6.2	4.8.2

4.5.2 Group B inspection. Assemblies which have passed the inspections of Group A shall be formed into lots (see 6.3). From these lots, sample assemblies shall be selected and subjected to the inspections of Table V. Sampling shall be double sampling in accordance with MIL-STD-105, AQL of 1.5 percent General Inspection Level II. If required, second sample testing shall be limited to those attributes failed during the first samples' testing.

TABLE V. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Brightness gain	3.6.3	4.8.3
Diopter focus	3.6.6	4.8.6
Focus adjustment	3.6.5	4.8.5
Collimation	3.6.4	4.8.4

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4.5.3 Group C inspections. Group C inspections shall be performed on assemblies that have passed Group A, or Group A and B inspections. Samples shall be randomly selected as follows and shall be subjected to inspections listed in Table VI.

4.5.3.1 Subgroup I. Samples selected for inspections listed in Table VI Subgroup I shall have passed Group A or Group A and B inspections. The sample size shall be in accordance with the following:

Production Quantity/Mo.	Sample Size	When Selected
3-9	1	At completion of 2/3 monthly production quantity
10-50	1	Selected from first ten units of monthly production quantity
51 or greater	1 per 50 units or any part thereof	Total number of samples selected from one-half the total monthly production quantity, or, one per fifty selected as for ten to fifty production quantity.

Tests may be performed in any order except immersion shall be performed last.

4.5.3.2 Subgroup II. One assembly that has passed Group A inspection shall be randomly selected every 12 months and subjected to the inspections listed in Table VI, Subgroup II.

4.5.3.3 Group C failures. Actions required in the event of a Group C failure shall be as specified in the contract or purchase order (see 6.2).

4.5.3.4 Disposition of Group C inspection units. Assemblies that have been subjected to the inspections listed in Table VI shall be refurbished as necessary and subjected to the controls and adjustments inspection of paragraph 4.8.1 as a minimum. If refurbishment requires disassembly, the resolution (4.8.2), diopter focus (4.8.6) and focus adjustment (4.8.5) inspections shall be performed. Failure to pass these requirements shall be cause for rejection of that assembly.

TABLE VI. Group C inspections

Inspections	Subgroup I	Subgroup II	Requirement paragraph	Test paragraph
Vibration	X		3.6.7	4.8.7
Shock	X		3.6.8	4.8.8
Temperature, humidity, altitude	X		3.6.9	4.8.9
Immersion	X		3.6.10	4.8.10
Arctic adapter assembly	X		3.5.1	4.7.1
Electromagnetic interference		X	3.6.11	4.8.11

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4.5.4 Group D inspection. This inspection shall consist of tests specified in Table VII and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE VII. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.10	4.9

4.5.4.1 Sampling for Group D inspection. Five assemblies shall be selected at random from the first article units for first article group D inspection. For subsequent quality conformance group D inspection, two assemblies shall be selected at random from every two month production lots.

4.5.4.2 Non-compliance. In case of failure of Group D inspection, procedures and requirements specified elsewhere in the contract shall be followed (see 6.2).

4.5.4.3 Disposition of Group D inspection units. Disposition of assemblies subjected to the inspection of Table VII shall be specified in the contract or purchase order (see 6.2).

4.6 Test methods. Test shall be conducted in accordance with the test procedures specified herein. Unless otherwise specified, the following conditions shall apply:

a. The radiation source used in the tests shall be a tungsten filament lamp operated at a color temperature of 2856 kelvin (K), ±50 K.
b. The photometer used for screen brightness measurements shall be a Pritchard Model 1970 PR or equal.
c. The photometer used for brightness measurements shall be calibrated against a standard source as specified below:
   (1) Tungsten filament lamp operated in conjunction with an opal glass such that the color temperature of the radiation emitted from the opal glass is 2856 K, ±50K.
   (2) Corning spectral filters Nos. 3-71 and 4-67 or equal.
   (3) Opal glass to produce a uniform, lambertian distribution.
   (4) Output brightness to be 0.1 to 1.0 footlambert uniformly distributed.
d. Tolerances on specified radiation levels shall be ± 10 percent.
e. Meters used for monitoring lamp current and voltage shall have an accuracy of ± 0.25 percent.
f. Neutral density filters used in test equipment shall have transmission characteristics within 10 percent of the nominal filter transmission from 0.35 to 1.0 micrometer.

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g. Tests shall be performed at ambient conditions as defined in 6.3 except where otherwise specified.
h. Test chambers used for environmental temperature tests shall maintain the temperature within $\pm 1.4^{\circ}C$.
i. Unless otherwise specified, tests shall be performed using hard source power supply. Tolerance on specified hard source voltages shall be within ± 0.05 volts direct current (Vdc).

4.7 Major components.

4.7.1 Arctic adapter assembly. The arctic adapter assembly shall be attached to an assembly using either a hard source or a BA-1567/U battery. Failure of the assembly to produce the characteristic green glow of the intensifier thru the eyepiece, or failure of the arctic adapter to meet the requirements of 3.5.1 shall constitute failure of this test.

4.7.2 Demist shield.

4.7.2.1 Fogging. Cool the demist shield to $0^{\circ}C$, $\pm 2^{\circ}C$ and then transfer it within 10 seconds to an atmosphere of 95 percent, ± 5 percent relative humidity and temperature of $33^{\circ}C$, $\pm 10^{\circ}C$. After 10 seconds, visually inspect the demist shields for fogging on the glass surface. Failure to meet the requirements of 3.5.2.1 shall constitute failure of this test.

4.7.2.2 Damage. The shield shall be tested in accordance with Method 518, Procedure I of MIL-STD-810. In step 3 the temperature shall be held at $-54^{\circ}C$ for one hour. In step 12, the temperature shall be $0^{\circ}C$. Delete steps 13, 14 and 15 and perform fogging test in accordance with 4.7.2.1. After completion of the fogging test, examine the demist shield for damage. Failure to meet the requirements of 3.5.2.2 shall constitute failure of this test.

4.8 Operational and environmental. All environmental tests requiring a performance check may have the check conducted with a battery BA-1567/U per MIL-B-18 as the power source. Each assembly shall demonstrate its ability to perform properly after being subjected to all the environmental tests (see 6.3) slated for it, but prior to any disassembly of the assembly. For the purpose of post environmental testing, "perform properly" shall be defined as the ability to successfuly complete the brightness gain (4.8.3), resolution (4.8.2) and collimation (4.8.4) tests.

4.8.1 Controls and adjustments. The assembly shall be examined for compliance with the requirements of 3.6.1. Failure of the assembly to meet the requirements of 3.6.1 shall constitute failure of the test.

4.8.2 Resolution. Place a 100 percent contrast, USAF 1951 Resolving Power Test Target (black on a clear background) in the focal plane of a collimator having a known focal length, back illuminate the target with 2856 K white light of such intensity, as to obtain maximum system resolution. Place the assembly with the range focus set at the infinity stop looking into the collimator. Determine the smallest pattern which can be resolved. Repeat this

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test with range focus set at the true infinity position. With the range focus set at the close focus stop, set the resolution target at a distance of not more than 25 cm from the assembly, and determine the smallest pattern which can be resolved. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.8.3 Brightness gain. Place a 2856 K extended lambertian light source at least 25 cm in front of the assembly objective. This extended lambertian source must be large enough to represent a 40° field-of-view to the assembly (i.e., the entire photocathode format is illuminated). Using a calibrated photometer, the extended light source shall be set to a brightness of not more than 5×10^{-4} foot lamberts. Activate the assembly looking at the extended source and read output through the eyepiece using a photometer calibrated to read correctly with a 5 mm entrance pupil. This 5 mm entrance pupil shall be placed on the optical axis of the photometer and the optical axis of the assembly at the specified 15 mm eye relief distance of the assembly. Gain is calculated as:

$$Gain = \frac{\text{Output brightness}}{\text{Input brightness}}$$

Failure to meet the requirement of 3.6.3 shall constitute failure of this test.

4.8.4 Collimation. Using equipment as described in Figure 1, 2 and 3 or equivalent, measure amount of decollimation in vertical and horizontal directions at system output, or assure (by use of Government approved GO-NO-GO equipment and procedure) that decollimation is within limits as specified in paragraph 3.6.4. Failure to meet the requirements of 3.6.4 shall constitute failure of this test.

4.8.5 Focus adjustment. The assembly focus range adjustments on each objective lens shall be rotated not less than 3 times throughout the total range of movement, and returned to the close focus (25 mm) position stop. A suitable torque measuring device shall be used to determine the break away torque of the focus range adjustments. Failure to meet the requirements of 3.6.5 shall constitute failure of the test.

4.8.6 Diopter focus. Measure the diopter adjustment of each eyepiece at each extreme of travel using a diopter scope whose entrance aperture is set at the eye relief distance. Rotate each eyepiece through its full adjustment range of not less than 3 times. Measure the torque from the objective end of each monocular via the diopter adjustment ring over full range of travel in each direction. Failure to meet the requirements of 3.6.6 shall constitute failure of this test.

4.8.7 Vibration. In part 1 of this test, the assembly shall be operated in the IR mode; the assembly shall be mounted to a test fixture and subjected to harmonic motion applied in each of three orthogonal planes; the frequency of the harmonic motion shall be varied uniformly from 10 to 55 Hz and return to 10 Hz in one minute; the amplitude shall be 0.05 inch, $\pm$ 0.005 inch double amplitude; this frequency sweep shall be repeated 10 times in each plane. In

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part 2 of this test the AN/PVS-5A shall be tested in accordance with Method 514.2, Procedure XI of MIL-STD-810. Failure to meet the requirements of 3.6.7 shall constitute failure of this test. Dents and finish damage to the shipping case and inserts shall not constitute failure, providing the protective function of the case is not impaired. Cases with such damage may be reused for tests.

4.8.8 Shock. The assembly in its carrying case and in turn, in its shipping/storage case shall be subjected to Test Method 516.2, Procedure II of MIL-STD-810. Failure to meet the requirements of 3.6.8 shall constitute failure of this test. Shipping case dents or finish damage shall not constitute failure providing the protective function of the case is not impaired. Cases with such damage may be reused for shock (drop) testing.

4.8.9 Temperature, humidity, altitude. The assembly shall be tested in accordance with Method 518.1 of MIL-STD-810 with the following variations: (1) During all cycles the temperature shall be -57°C in steps 2 and 3, and steps 12 and 13; in step 5 the time shall be 2 hours; in step 7 the time shall be 5 hours; and in step 8, when the temperature reaches +45°C it shall be held constant for 1-1/2 hours. (2) During the first cycle the high altitude sequence shall be performed; the chamber pressure shall be increased to the equivalent of 10,000 feet above sea level and the temperature increased to -45°C between steps 3 and 4. In this environment the assembly shall be operated for one hour and an operational check performed. The lens shall be normally exposed on the first cycle only. The maximum temperature in steps 6 and 7 shall be +49°C. In step 8 an operational check shall be made when the temperature reaches +45°C. The time for completion of step 8 is 8 hours, including the 1-1/2 hour pause. Upon completion of step 9 return pressure to ambient conditions (step 11). (3) During cycles 2, 3, and 4 the pressure shall remain at ambient level and an operational check shall be performed. The time for completion of step 8 is 10 hours including the 1-1/2 hour pause. The temperature shock test shall be performed separately in accordance with Method 503.1 of MIL-STD-810 except that the temperature extremes shall be +65°C and -57°C and the assembly shall be held at the temperature extremes for 2 hours in each step instead of four hours. A maximum interval of two (2) hours at room temperature is permissible in lieu of the 5 minute limitation as described in steps 2 and 4 of Method 503.1 of MIL-STD-810. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

4.8.10 Immersion. The assembly less the face cushion and the head straps shall be subjected to Test Method 512.1, Procedure I of MIL-STD-810 except that the immersion time shall be 30 minutes minimum. Any evidence of water leakage shall constitute failure of this test.

4.8.11 Electromagnetic interference. The assembly shall be tested for the requirements specified in 3.6.11 in conformance with test methods of MIL-STD-462. Failure to meet the requirements of 3.6.11 shall constitute failure of this test.

4.9 Reliability. The reliability test shall be run unless otherwise stated in the contract or purchase order (see 6.2).

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4.9.1 Reliability test. The test shall be performed under the test conditions specified herein. The reliability test shall be conducted in accordance with test level B of MIL-STD-781. ON-OFF cycling shall consist of 55 minutes of operation (ON time) followed by 5 minutes OFF. Input voltage to the assembly shall be 2.7 Vdc. The initial brightness gain of each tube assembly shall be between 7,500 and 15,000. Acceptable limits of brightness gain during and at the completion of the test shall be 5,000 to 20,000. No gain adjustments shall be allowed during the test or inspections. The signal-to-noise ratio (S/N) shall be greater than 3.20 throughout the test and inspections. Operation shall be with a target located 20 inches from the front of the objective lens with the system focused on the target. The target shall have a color temperature of 2856 K and completely fill the field of view of the system under test. The target brightness shall be 1×10^{-4} footlamberts which shall be raised to 5×10^{-3} footlamberts for 5 seconds and then to 50 footlamberts for 3 seconds once during each "ON" period. Sufficient instrumentation shall be provided to ensure immediate recognition of a catastrophic failure as well as a change in relative monocular assembly output brightness. System performance shall be measured at least once each 200 hours of assembly operation time by performing at least the resolution, collimation, and brightness gain inspections. During the period that system performance is being measured, each of the following controls and adjustments shall be actuated through the full adjustment range 10 times: rotary switch (ON-OFF-IR), eye relief and tilt, interpupilary adjustment, objective lens focus and eyepiece diopter adjustment. Image intensifier tube assembly parameters luminance gain, EBI, signal-to-noise, output brightness uniformity, and photocathode, microchannel plate and screen quality shall be tested according to MIL-I-49052 not less than once every 200 hours of assembly operating time. A failure shall be presumed to have occurred immediately after the last successful measurement of inspection. An assembly shall be removed from the test for failure analysis upon recognition of a failure. If the failure is in an image intensifier tube assembly, the failed image intensifier tube assembly shall be removed from the test and the assembly returned to the test except when both image intensifier tube assemblies have been removed from test. All other failures on the assembly shall be repaired and the unit returned to test. Upon removal of a tube assembly from test, the tube assembly shall be considered failed for the purpose of evaluating assembly reliability. A failed tube assembly, repaired and returned to test shall be for information only. Each assembly and image intensifier tube assembly on test at the completion of the test shall be submitted to at least the Group A and Group B inspections performed during the test. The following requirements of MIL-STD-781 shall apply:

a. Design and performance tests
b. Detail reliability test procedures
c. Inspection
d. Test facilities
e. Selection of equipment
f. Installation of equipment
g. Testing the set-up
h. Failure categories
i. Verifying repair
i. Verifying repair.

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4.9.2 Failure definition. A failure is defined as the inability of the assembly and/or tube assembly(s) to meet the requirements of Group A and Group B inspections. This includes the inability of the tube assembly to meet the Group A and Group B inspections of MIL-I-49052.

4.9.3 Accept/reject criteria.

4.9.3.1 First article reliability. The accept/reject determination shall be made upon either the accumulation of 18,400 hours total operation time on the tube assemblies or the occurrence of 6 assembly failures. Only operating time to first failure will be used for any or all tube assemblies which have experienced a failure. Six or more assembly failures shall constitute failure of the equipment to meet Group D inspection requirements. Five or less assembly failures shall constitute acceptance of the Group D inspection.

4.9.3.2 Quality conformance reliability. Each assembly on test shall be operated until a total operating "ON" time of 2000 hours is accumulated on each tube assembly. If a tube assembly fails before the accumulated 2000 hours, the failed tube assembly will be counted as a failure and the assembly minus the failed tube assembly shall be put back on test until the remaining tube assembly either fails or accumulates the required 2000 hours of "ON" time. Two or more assembly failures shall constitute failure of the equipment to meet Group D inspection requirements, one or less assembly failures shall constitute acceptance of Group D inspection.

4.9.4 Non-relevant failures. Any failure attributed directly and solely to over-stress induced by the inspection procedure, such as, a tube assembly connecting wire breaking after the first 200 hour inspection point, shall be neither regarded as relevant in the accept/reject determination, nor cause for removal from the test, if the failure is repairable. Any non-relevant failure of this type which is non-repairable shall require the replacement of such image intensifier assembly(s). The time on the replacement shall begin at zero. The time on the removed assembly(s) shall not be used in the accept/reject determination of either the assembly or image intensifier tube assemblies. The contractor shall have the option to classify any non-relevant failure to be relevant.

4.10 Inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance to requirements of Section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Night Vision Goggles, AN/PVS-5A shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Night Vision Goggles, AN/PVS-5A shall be dried in accordance with the applicable procedures of MIL-P-116.

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5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Night Vision Goggles, AN/PVS-5A. Each item shall be individually unit packed method III as follows: place each major assembly into carrying case, close cover and secure fasteners provided. Place the carrying case and accessories within the transit case. Position the cover and secure the latches. Cushion the transit case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place each cushioned case within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.(2)). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Night Vision Goggles, AN/PVS-5A unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of Night Vision Goggles, AN/PVS-5A unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds.

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Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The AN/PVS-5A is intended for general purpose viewing during periods of darkness.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.
b. The quantities for each of the following major components:
   Demist shield assembly per SM-C-657428
   Arctic adapter assembly per SM-DL-567420
c. That the contractor will furnish sufficient BA 1567/U batteries per MIL-B-18 for all tests. The batteries used for testing are not to be delivered.
d. Necessary actions by the contractor in the event of a Group C (4.5.3.3) or Group D (4.5.4.2) failure.
e. Disposition of Group D inspection units (4.5.4.3).
f. Level A, level B or Commercial preservation and packing (see Section 5).
g. If rough handling and functional test are required.
h. Batteries will not be supplied with the assembly. Prior to packing and packaging, each assembly shall be examined to verify absence of batteries.
i. Quantity and schedule for First Article testing for all line items.
j. Whether first article requirements for the tube are required for AN/PVS-5A systems bought under this specification.

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k. That when Image Intensifier tubes are obtained as part of the Goggle assembly reliability shall be performed in accordance with MIL-I-49052 rather than in accordance with MIL-N-49065. When Goggle assemblies are obtained less Image Intensifier tube assemblies, the reliability test of this specification shall be waived.

l. Environmental pollution prevention measures are contained in the packing material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.3 Definitions.

6.3.1 Damage. Damage is defined as an electrical failure of malfunctioning including arcing, corona, intermittent operation, flickering, brittleness, cracking, corrosion or failure of any finish, condensation, fogging or moisture on any internal surface; dents, or breakage of any hardware, connection, or component; or any change in binocular alignment.

6.3.2 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

6.3.3 Inspection lot size. An inspection lot is defined as not more than those assemblies successfully completing Group A tests each month.

6.3.4 Environmental tests. For the purpose of this specification, environmental tests are those tests discussed in 4.8.7 through 4.8.10.

6.3.5 Room temperature. Room temperature is defined as +23°C, ±3°C.

6.3.6 Degradation. A significant change in measurable characteristics which results in the assembly no longer meeting specified requirements or indicates that there is an inherent defect in the operating characteristic of the unit.

6.4 International standardization agreement. Provisions of this publication are the subject of an international standardization agreement (QSTAG 460). When amendment, revision, or cancellation of this publication is proposed that will affect or violate the agreement concerned, the preparing activity will take proper action through international standardization channels. If local office is not available contact Commander, US Army Materiel Development & Readiness Command, ATTN: DRCIRD, 5001 Eisenhower Avenue, Alexandria, VA 22333.

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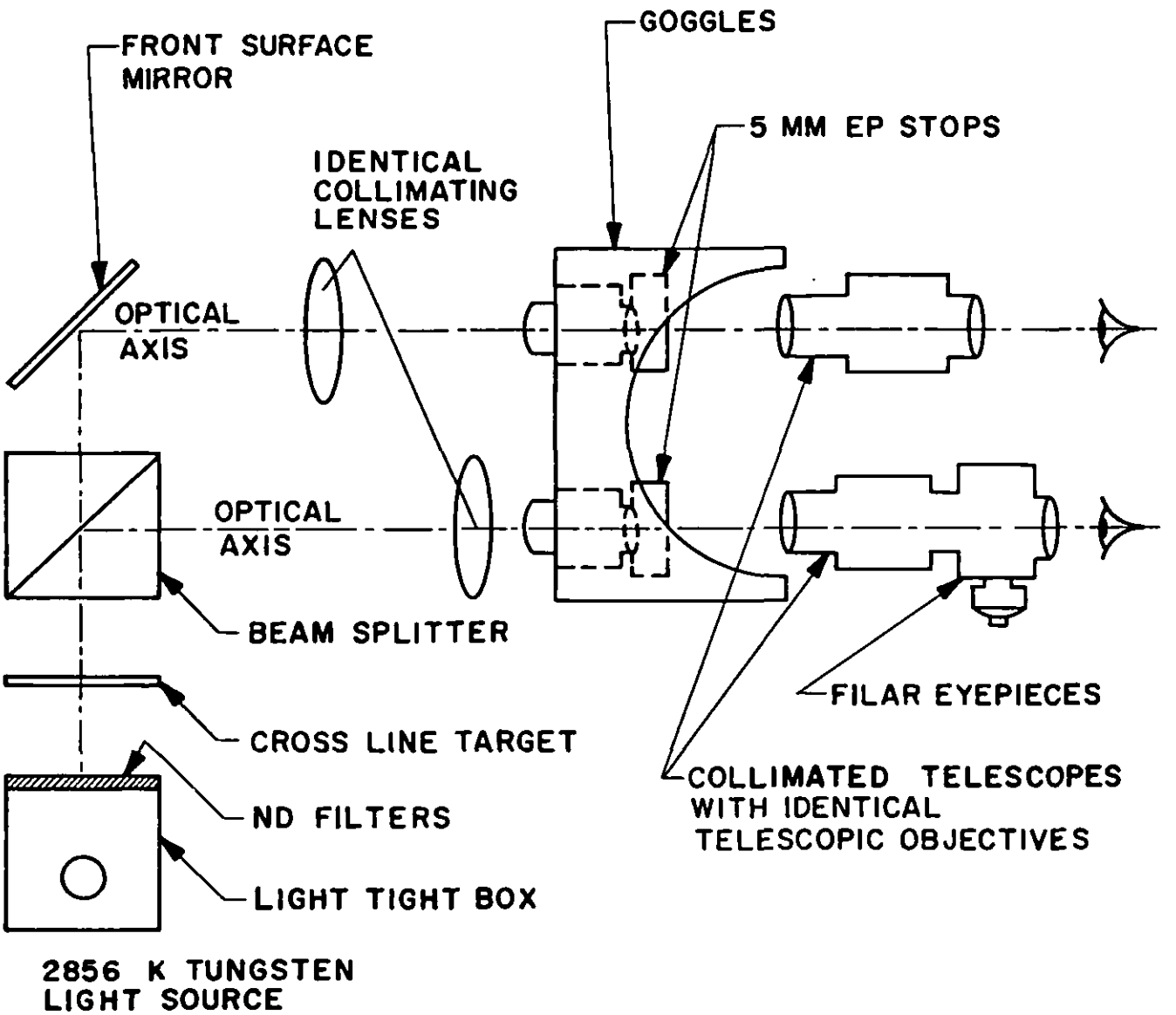


FIGURE 1. ASSEMBLY COLLIMATION

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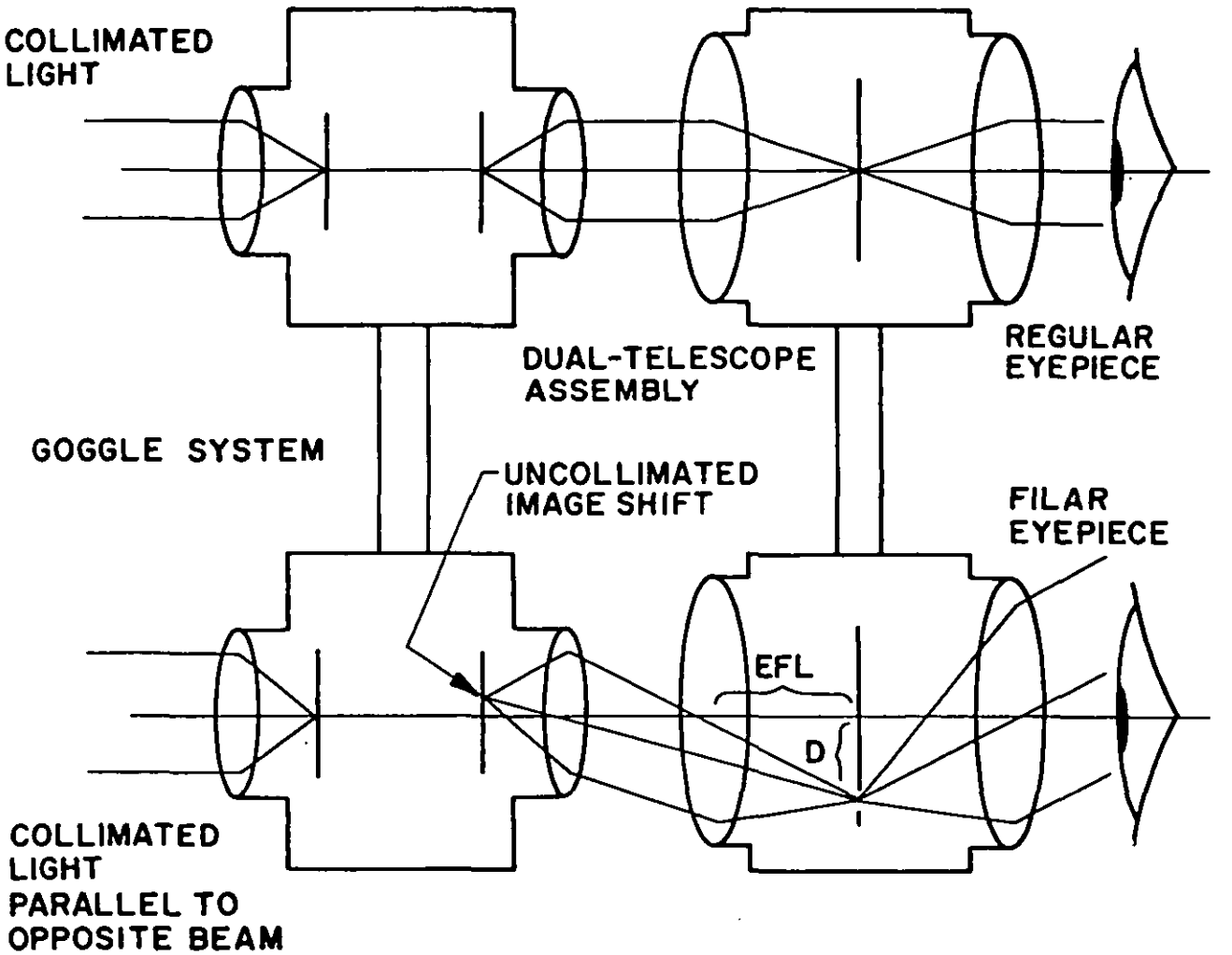
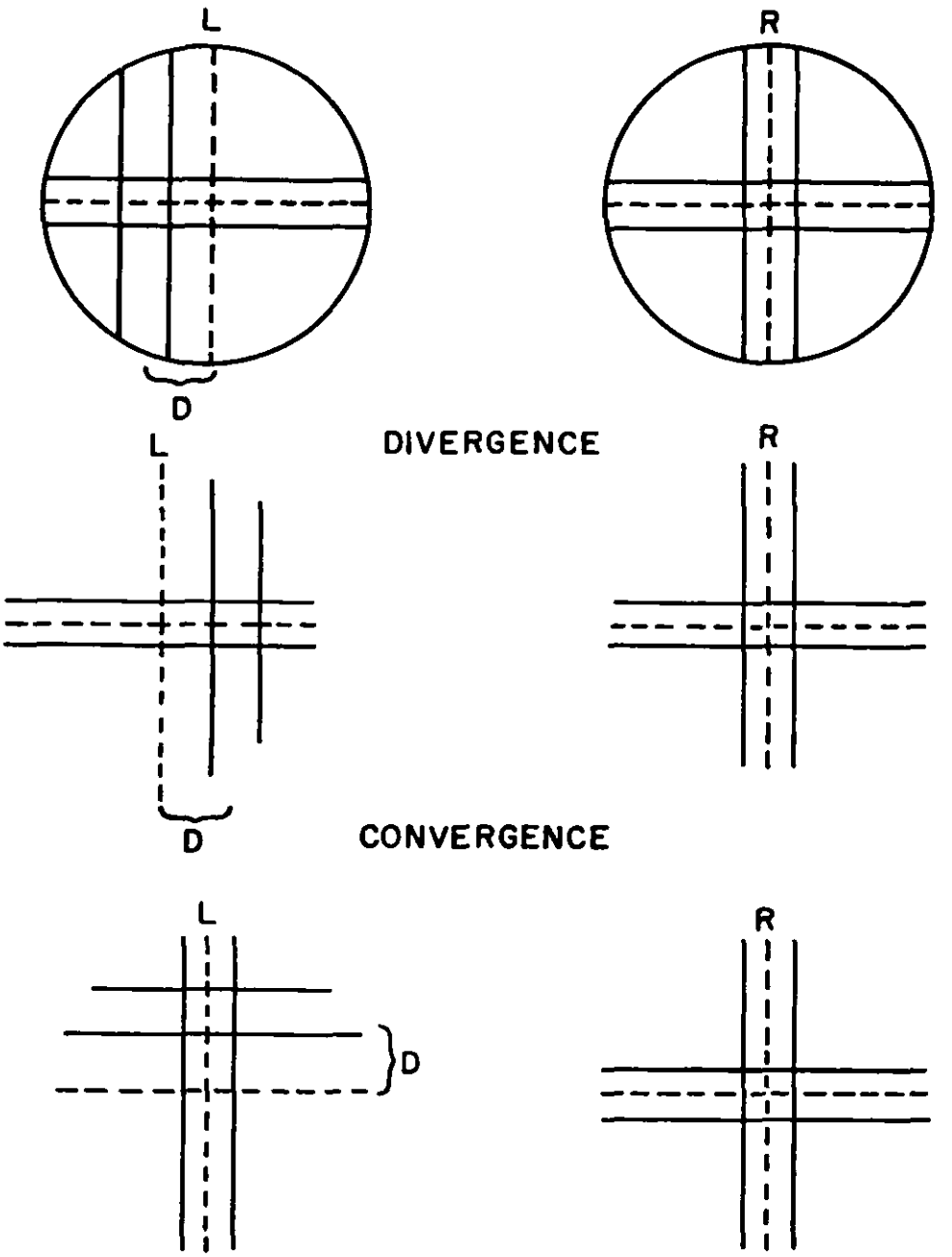


FIGURE 2. ASSEMBLY COLLIMATION

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$$\text{TAN } \theta = \left(\frac{D}{EFL}\right)$$

$$\text{THUS } \theta = \text{TAN}^{-1} \left(\frac{D}{EFL}\right)$$

FIGURE 3. ASSEMBLY COLLIMATION

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